

29	Answer: C
	Paper stops alpha particles, so $\alpha + \beta + \text{background} = 352\text{min}^{-1}$ $\beta + \text{background} = 256\text{min}^{-1}$ Initial count of beta particle source = $256 - 16 = 240\text{min}^{-1}$ Initial count of alpha particle source = $352 - 256 = 96\text{min}^{-1}$ After 12 days, alpha particle source undergoes 3 half-lives and beta particle source undergoes 4 half-lives Final count of alpha particle source = $\frac{96}{2^3} = 12\text{min}^{-1}$ Final count of beta particle source = $\frac{240}{2^4} = 15\text{min}^{-1}$ Total count = $12 + 15 + 16 = 43\text{min}^{-1}$

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